



JAIPUR NATIONAL UNIVERSITY

End Semester/Year Examination Dec-2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech(Computer Science Engineering / AIML / CY / Data Science)
Semester/Year:	5/3
Paper Code:	BTECSCO501T22
Title of Paper:	Software Engineering
Time: 3:00 Hour	Max. Marks: 60

INSTRUCTIONS

1. This question paper consists of two sections, Section-A and Section-B.
 2. Section-A consists of seven questions and
 3. Section-B consists of six questions.
- Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION - A	Marks Allotted	Bloom Level	Course Outcome
Q.1	Define the term software engineering .	4	1	1
Q.2	Explain in your own words why the software development life cycle (SDLC) is important in software engineering.	4	2	1
Q.3	Given the requirement "When a user logs in, the system shall display the user's dashboard within 3 seconds", identify whether it is a functional requirement or a non-functional requirement, and justify your choice.	4	3	2
Q.4	Explain in your own words why a well-written SRS document reduces the risk	4	2	2

of misunderstandings between stakeholders and developers.

Q.5	Discuss Top Down Approach of Software development.	4	3	3
Q.6	Analyze the differences between White-box and Black-Box Testing.	4	4	4
Q.7	Explain the need of Software Maintenance.	4	3	5

SECTION – B

Q.1	Define Software Engineering. Explain in detail the Software Development Life Cycle (SDLC) and discuss its different phases with suitable examples.	10	1,2	1
Q.2	Explain the need and importance of a Software Requirement Specification document. Describe the structure and major components of an SRS with suitable examples.	10	2	2
Q.3	Evaluate the qualities of a good SRS. Discuss how incomplete or ambiguous requirements can affect software project success.	10	5	2
Q.4	Analyze the role of modularity, abstraction, and cohesion in effective software design.	10	4	3
Q.5	Compare white-box and black-box testing techniques. Evaluate their applicability to different testing scenarios	10	4	4
Q.6	Define software maintenance. Discuss its types (corrective, adaptive, perfective, and preventive) with suitable examples.	10	2	5